

Mathematical Analysis II

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Contents

Preface	iv
1 Sequences and Series of Functions	1
1.1 Discussion of Main Problem	1
1.2 Uniform Convergence	1
1.3 Uniform Convergence and Continuity	1
1.4 Uniform Convergence and Integration	1
1.5 Uniform Convergence and Differentiation	1
1.6 Equicontinuous Families of Functions	1
1.7 The Stone-Weierstrass Theorem	1
2 Some Special Functions	2
2.1 Power Series	2
2.2 The Exponential and Logarithmic Functions	2
2.3 The Trigonometric Functions	2
2.4 The Algebraic Completeness of the Complex Field	2
2.5 Fourier Series	2
2.6 The Gamma Function	2
3 Functions of Several Variables	3
3.1 Linear Transformations	3
3.2 Differentiation	3
3.3 The Contraction Principle	3
3.4 The Inverse Function Theorem	3
3.5 The Implicit Function Theorem	3
3.6 The Rank Theorem	3
3.7 Determinants	3
3.8 Derivatives of Higher Order	3
3.9 Differentiation of Integrals	3
4 Integration of Differential Forms	4
4.1 Integration	4
4.2 Primitive Mappings	4

4.3	Partitions of Unity	4
4.4	Change of Variables	4
4.5	Differential Forms	4
4.6	Simplexes and Chains	4
4.7	Stokes' Theorem	4
4.8	Closed Forms and Exact Forms	4
4.9	Vector Analysis	4
5	The Lebesgue Theory	5
5.1	Set Functions	5
5.2	Construction of the Lebesgue Measure	5
5.3	Measure Spaces	5
5.4	Measurable Functions	5
5.5	Simple Functions	5
5.6	Integration	5
5.7	Comparison with the Riemann Integral	5
5.8	Integration of Complex Functions	5
5.9	Functions of Class $\mathcal{L}^2$	5

Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \oplus > \vee > \implies, \iff > \iff$$

Hence

$$P \oplus Q \implies \neg R \vee S \wedge T \iff U \iff V$$

is understood as

$$((P \oplus Q) \implies ((\neg R) \vee (S \wedge T))) \iff (U \iff V)$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

merely expresses that P , and therefore Q in the current discussion; it is not itself a proposition.

Chapter 1

Sequences and Series of Functions

1.1 Discussion of Main Problem

1.2 Uniform Convergence

1.3 Uniform Convergence and Continuity

1.4 Uniform Convergence and Integration

1.5 Uniform Convergence and Differentiation

1.6 Equicontinuous Families of Functions

1.7 The Stone-Weierstrass Theorem

Chapter 2

Some Special Functions

2.1 Power Series

2.2 The Exponential and Logarithmic Functions

2.3 The Trigonometric Functions

2.4 The Algebraic Completeness of the Complex Field

2.5 Fourier Series

2.6 The Gamma Function

Chapter 3

Functions of Several Variables

3.1 Linear Transformations

3.2 Differentiation

3.3 The Contraction Principle

3.4 The Inverse Function Theorem

3.5 The Implicit Function Theorem

3.6 The Rank Theorem

3.7 Determinants

3.8 Derivatives of Higher Order

3.9 Differentiation of Integrals

Chapter 4

Integration of Differential Forms

4.1 Integration

4.2 Primitive Mappings

4.3 Partitions of Unity

4.4 Change of Variables

4.5 Differential Forms

4.6 Simplexes and Chains

4.7 Stokes' Theorem

4.8 Closed Forms and Exact Forms

4.9 Vector Analysis

Chapter 5

The Lebesgue Theory

5.1 Set Functions

5.2 Construction of the Lebesgue Measure

5.3 Measure Spaces

5.4 Measurable Functions

5.5 Simple Functions

5.6 Integration

5.7 Comparison with the Riemann Integral

5.8 Integration of Complex Functions

5.9 Functions of Class $\mathcal{L}^2$

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